

Organization

About Me:

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- Whatsapp Group



Study Centre:

- Each Wednesday 18:00-20:00, ETF C1
- We will try to answer all your dynamics related questions in a peer-to-peer discussion

Midterm:

- There is a voluntary midterm exam that will be on [moodle](#)

Stack:

- 3-4 Moodle STACK Questions will be posted each week on Wednesday after class
- Completion is voluntary but successful completion can yield up to a [0.25 grade bonus](#)
- The [questions are available for seven days](#) but you have only [one chance](#)

Theory Week 1

no equilibrium

Whatsapp Group

Mechanics 3 CAB G59
WhatsApp group



1. Single particle Dynamics

Def. particle: A **particle** is an idealized view of a (small) object whose size and shape have a negligibly influence on its motion. Of course, no object is negligibly small in reality. However, when the object's shape

↳ We assume: • Mass concentrated in one point
• only Translation (no rotation)

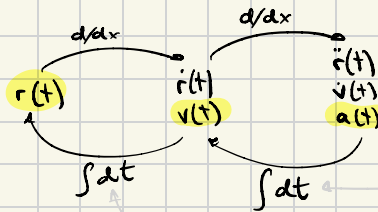
Def. kinematics & kinetics: of freedom. In the following, we will focus on a single particle and study its motion through space (which we call its **kinematics**), and we will discuss how its motion depends on the forces acting on the particle (which we call the **kinetics**).

1.1 Kinematics

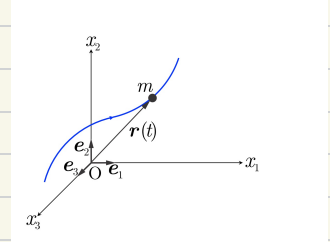
□ **position:** $\underline{r} = \sum_{i=1}^3 x_i \underline{e}_i = x_1 \underline{e}_1 + x_2 \underline{e}_2 + x_3 \underline{e}_3$

□ **degrees of freedom (DOFs):** a particle in **d** dimensions has **d** DOFs

□ **General Motion:**

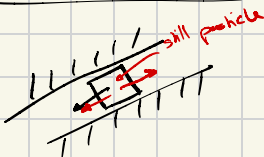


time-invariant!! = inertial frame (weak for us)
Fixed Cartesian reference frame



don't forget
integration constant!

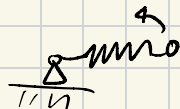
Constrained Motion



$$\underline{d - k = \text{DOF's}} \rightarrow \underline{n \cdot d - \underline{r} = \text{DOF's}}$$



1 rod = 1 constraint (most of the time)



springs = no constraints